Example #1

- Sequence: $n = 3$ (3 tokens). • key/query dimension: $d_K = 2$
- Value dimension: $d_V = 2$

$$\text{Attention}(Q,K,V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_K}}\right)V$$

1. Define Q, K, V explicitly
   - Think of each row as a token's representation (row-vector view)

Let:

$$Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix}, \quad K = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \quad V = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 3 & 1 \end{bmatrix}$$

Interpretation:

- Token 1: $q_1 = [1\ 0]$, $k_1 = [1\ 0]$, $V_1 = [1\ 0]$
- Token 2: $q_2 = [0\ 1]$, $k_2 = [1\ 1]$, $V_2 = [0\ 2]$
- Token 3: $q_3 = [1, 1]$, $k_3 = [0\ 1]$, $V_3 = [3\ 1]$

2. Compute raw scores: QK^T

$$K = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \longrightarrow T \longrightarrow K^T = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}$$

$$QK^T = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} =$$

Q_1K^T

$Q_1 = [1\ 0]$ $K_1 = [1, 0] \rightarrow Q_1K_1^T = [1, 0][1, 0] = 1 \cdot 1 + 0 \cdot 0 = 1$

$Q_1 \cdot K_2^T = [1, 0][1, 1] \rightarrow 1 \longrightarrow \text{row } 1 \rightarrow Q_1K^T = [1\ 1\ 0]$

Q_2K^T

$Q_2 = [0, 1]$ $K_1 = [1\ 0] \rightarrow Q_2K_1^T = [0\ 1][1\ 0] = 0$

$Q_2 = [0, 1]$ $K_2 = [1\ 1] \rightarrow Q_2K_2^T = [0, 1][1\ 1] = 1$ $\rightarrow Q_2K^T = [0, 1, 1]$

$Q_2 = [0, 1]$ $K_3 = [1\ 0] \rightarrow Q_2K_3^T = [0, 1][0\ 1] = 1$

$Q_3K^T = [1\ 1]\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} = [1, 2, 1] \longrightarrow Q_3K^T = [1\ 2\ 1]$

$$QK^T = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix}$$

Entry (i, j) is how much query i matches key j